

Chapter 05: Diffusion in Solids & Industrial Carburization

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Materials Science (MIAE 221)

Textbook:

Materials Science and Engineering: An Introduction

(10th Edition) by William D. Callister, Jr. & David G. Rethwisch — Chapter 5

1. Executive Overview & First-Principles Philosophy

In crystalline solids at room temperature, matter appears completely rigid and stationary. Yet at the atomic scale, atoms vibrate incessantly around their equilibrium lattice positions at frequencies of $\approx 10^{13}$ Hz. Given sufficient thermal energy, atoms overcome potential energy barriers and hop from site to site.

Diffusion is the macroscopic phenomenon of material transport by atomic motion. It is the fundamental kinetic mechanism underpinning virtually all heat treatments and manufacturing processes in engineering:

- **Surface Hardening (Carburization & Nitriding):** Infusing carbon or nitrogen atoms into the surface of low-carbon steel gears to create an ultra-hard, wear-resistant outer casing while preserving a shock-absorbing, ductile core.
 - **Microelectronic Semiconductor Doping:** Introducing phosphorus or boron atoms into ultra-pure silicon wafers at nanometer depths to create $p - n$ junctions in microprocessors.
 - **Sintering of Ceramic Powders:** Consolidating compressed ceramic powder into dense, pore-free components without melting.
 - **Precipitation Hardening & Age Hardening:** Precipitating nanoscale intermetallic phases in aerospace aluminum alloys (Al 7075, Al 2024) to lock dislocations and quadruple yield strength.
 - **Creep Rupture in Jet Engines:** High-temperature deformation governed by vacancy-assisted atomic diffusion under continuous centrifugal stress.
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2. Diffusion Mechanisms: How Atoms Migrate (Callister §5.2)

Solid-state diffusion is divided into two broad categories:

1. **Self-Diffusion:** The migration of host atoms within a pure elemental metal (e.g., radioactive ^{64}Cu tracer atoms diffusing through a block of pure copper). There is no net chemical composition change.
2. **Interdiffusion (Impurity Diffusion):** The migration of solute atoms through a solvent matrix driven by a concentration gradient (e.g., copper and nickel atoms interdiffusing across a bonded interface).

(b) Note the kind of diffusion for which each of these equations is normally applied.

5.1 INTRODUCTION

diffusion

WileyPLUS

Tutorial Video:
What Is Diffusion?

Many reactions and processes that are important in the treatment of materials rely on the transfer of mass either within a specific solid (ordinarily on a microscopic level) or from a liquid, a gas, or another solid phase. This is necessarily accomplished by **diffusion**, the phenomenon of material transport by atomic motion. This chapter discusses the atomic mechanisms by which diffusion occurs, the mathematics of diffusion, and the influence of temperature and diffusing species on the rate of diffusion.

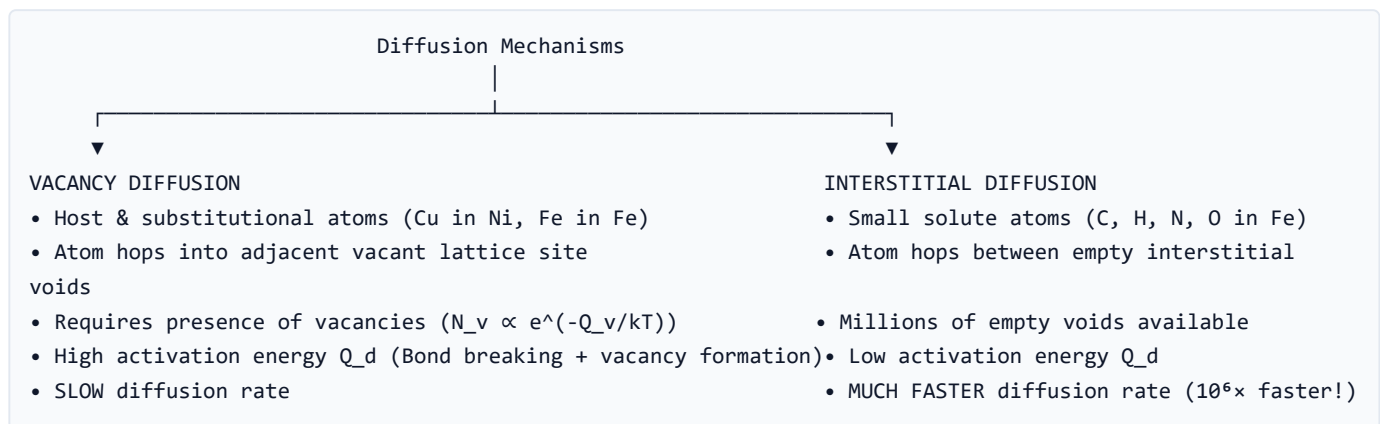
The phenomenon of diffusion may be demonstrated with the use of a *diffusion couple*, which is formed by joining bars of two different metals together so that there is intimate contact between the two faces; this is illustrated for copper and nickel in Figure 5.1a, which includes schematic representations of atom positions and composition across the interface. This couple is heated for an extended period at an elevated temperature (but below the melting temperature of both metals) and cooled to room temperature. Chemical analysis reveals a condition similar to that represented in Figure 5.1b—namely, pure copper and nickel at the two extremities of the couple, separated by an alloyed region. Concentrations of both metals vary with position as shown in Figure 5.1b (bottom). This result indicates that copper atoms have migrated or diffused into the nickel, and that nickel has diffused into copper. The process by which atoms of one metal diffuse into another is termed

interdiffusion

interdiffusion, or **immunity diffusion**.

Figure 5.1: (a) A diffusion couple formed by joining pure copper and pure nickel bars. (b) Atomic mixing across the interface after prolonged high-temperature exposure. (c) Resulting smooth concentration profile — from Callister & Rethwisch 10th Ed. (Fig. 5.1).

The Two Foundational Atomic Hopping Mechanisms:



A. Vacancy Diffusion

- **Physical Mechanism:** An atom residing on a normal lattice site jumps into an adjacent empty lattice vacancy.
- **Energetic Requirements:** For a jump to occur, two conditions must be satisfied simultaneously:
 1. An adjacent lattice site must be vacant (governed by vacancy formation energy Q_v).
 2. The atom must possess sufficient thermal vibrational energy to break bonds with its current neighbors and squeeze past surrounding lattice atoms into the vacancy (governed by migration energy Q_m).
- **Activation Energy:** The overall activation energy for vacancy diffusion is the sum of formation and migration energies:

$$Q_d = Q_v + Q_m$$

- Operates for all **self-diffusion** and **substitutional solid solutions** (e.g., Ni in Cu, Zn in Cu).

B. Interstitial Diffusion

- **Physical Mechanism:** Small solute atoms (Carbon, Nitrogen, Hydrogen, Oxygen) migrate directly from one interstitial void site to an adjacent empty interstitial site.
- **Why Interstitial Diffusion is Orders of Magnitude Faster than Vacancy Diffusion:**
 1. **Void Availability:** In any crystal lattice, virtually all interstitial void sites are unoccupied. An interstitial atom never has to wait for a vacancy to arrive; adjacent vacant sites are always available!
 2. **Lower Activation Energy:** Small interstitial atoms do not have to break host lattice bonds to move; they merely squeeze between host atoms, requiring much lower migration energy:

$$Q_{d,\text{interstitial}} \ll Q_{d,\text{vacancy}}$$

- *Dramatic Engineering Comparison*

: At 1000°C in iron:

- Diffusion coefficient of interstitial Carbon: $D_{\text{C}} \approx 2.4 \times 10^{-11} \text{ m}^2/\text{s}$
- Diffusion coefficient of self-diffusing Iron: $D_{\text{Fe}} \approx 2.3 \times 10^{-16} \text{ m}^2/\text{s}$
- **Carbon diffuses 100,000 times faster than iron at the exact same temperature!**

3. Steady-State Diffusion: Fick's First Law (Callister §5.3)

3.1 The Concept of Diffusion Flux (J)

The rate of mass transfer is quantified by the **Diffusion Flux (J)**: the mass (or number of atoms) M diffusing perpendicularly through a unit cross-sectional area A per unit time t :

$$J = \frac{M}{A \cdot t} = \frac{1}{A} \frac{dM}{dt}$$

- **SI Units:** $\text{kg}/(\text{m}^2 \cdot \text{s})$ or $\text{g}/(\text{cm}^2 \cdot \text{s})$ (or $\text{atoms}/(\text{m}^2 \cdot \text{s})$).

3.2 Fick's First Law

When conditions are **steady-state** (meaning the diffusion flux J and the concentration profile do not change with time, $\frac{\partial C}{\partial t} = 0$), mass transport is governed by **Fick's First Law**:

$$J = -D \frac{dC}{dx}$$

where:

- J = diffusion flux ($\text{kg}/(\text{m}^2 \cdot \text{s})$).
- D = **Diffusion Coefficient** (m^2/s or cm^2/s).
- $\frac{dC}{dx}$ = **Concentration Gradient** (kg/m^4 or g/cm^4):

$$\frac{dC}{dx} \approx \frac{\Delta C}{\Delta x} = \frac{C_{\text{low}} - C_{\text{high}}}{x_{\text{low}} - x_{\text{high}}}$$

- **The Negative Sign:** Reflects the physical reality that mass naturally diffuses **down the concentration gradient**—from regions of high chemical concentration to regions of low concentration. Because ΔC is negative in the direction of positive x , the negative sign ensures that the calculated mass flux J is a positive quantity.

4. Nonsteady-State Diffusion: Fick's Second Law & Carburization (Callister §5.4)

In virtually all practical engineering applications, diffusion is **nonsteady-state**: the concentration of the diffusing species at any spatial location changes dynamically over time:

$$C = C(x, t)$$

4.1 Fick's Second Law

Applying mass conservation across an infinitesimal volume element yields **Fick's Second Law**:

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right)$$

Assuming the diffusion coefficient D is independent of concentration across the alloy range:

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

where C_x represents the concentration at depth x after time t . The expression $\text{erf}(x/2\sqrt{Dt})$ is the Gaussian error function,⁴ values of which are given in mathematical tables for various $x/2\sqrt{Dt}$ values; a partial listing is given in Table 5.1. The concentration parameters that appear in Equation 5.5 are noted in Figure 5.5, a concentration profile taken at a specific time. Equation 5.5 thus demonstrates the relationship between concentration, position, and time—namely, that C_x , being a function of the dimensionless parameter $x/\sqrt{Dt}$, may be determined at any time and position if the parameters C_0 , C_s , and D are known.

Suppose that it is desired to achieve some specific concentration of solute, C_1 , in an alloy; the left-hand side of Equation 5.5 now becomes

$$\frac{C_1 - C_0}{C_s - C_0} = \text{constant}$$

considered to be semi-infinite if none of the diffusing atoms reaches the bar end during the time t takes place. A bar of length l is considered to be semi-infinite when $l > 10\sqrt{Dt}$.

The error function is defined by

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-y^2} dy$$

where y has been replaced by the variable z .

Figure 5.5: Nonsteady-state diffusion concentration profiles $C(x, t)$ at successive times $t_1 < t_2 < t_3$ into a semi-infinite solid with constant surface concentration C_s — from Callister & Rethwisch 10th Ed. (Fig. 5.5).

4.2 The Semi-Infinite Solid Boundary-Value Problem (Industrial Carburization)

Consider a steel bar placed in a carbon-rich gas furnace atmosphere. The mathematical model assumes a **semi-infinite solid**:

- **Initial Condition:** Before diffusion begins ($t = 0$), the solute is uniformly distributed at initial bulk concentration C_0 :

$$C(x, 0) = C_0 \quad \text{for all } 0 \leq x \leq \infty$$

- **Boundary Conditions:**

1. At the surface ($x = 0$), the concentration is instantaneously brought to and maintained at a constant surface concentration C_s by the furnace gas:

$$C(0, t) = C_s \quad \text{for all } t > 0$$

2. Infinitely deep into the interior ($x \rightarrow \infty$), the concentration remains unaffected:

$$C(\infty, t) = C_0 \quad \text{for all } t > 0$$

The Exact Analytical Solution:

Solving Fick's Second Law using Laplace transforms or similarity variables yields:

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

where:

- $C_x = C(x, t)$ = concentration at depth x after elapsed time t .
- C_s = constant surface concentration.
- C_0 = uniform initial bulk concentration.
- x = depth below surface (m).
- D = diffusion coefficient at furnace temperature (m^2/s).
- t = diffusion time (s).
- $\operatorname{erf}(z)$ = **Gaussian Error Function**, defined mathematically as:

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-y^2} dy$$

4.3 Tabulated Values of the Error Function & Linear Interpolation

z	$\text{erf}(z)$	z	$\text{erf}(z)$	z	$\text{erf}(z)$
0.00	0.0000	0.50	0.5205	1.10	0.8802
0.05	0.0564	0.55	0.5633	1.20	0.9103
0.10	0.1125	0.60	0.6039	1.30	0.9340
0.15	0.1680	0.65	0.6420	1.40	0.9523
0.20	0.2227	0.70	0.6778	1.50	0.9661
0.25	0.2763	0.75	0.7112	1.60	0.9763
0.30	0.3286	0.80	0.7421	1.70	0.9838
0.35	0.3794	0.85	0.7707	1.80	0.9891
0.40	0.4284	0.90	0.7969	1.90	0.9928
0.45	0.4755	0.95	0.8209	2.00	0.9953

Linear Interpolation Formula:

To find an argument z corresponding to an error function value $\text{erf}(z)$ lying between tabulated points $(z_1, \text{erf}(z_1))$ and $(z_2, \text{erf}(z_2))$:

$$\frac{z - z_1}{z_2 - z_1} = \frac{\text{erf}(z) - \text{erf}(z_1)}{\text{erf}(z_2) - \text{erf}(z_1)} \implies z = z_1 + (z_2 - z_1) \left[\frac{\text{erf}(z) - \text{erf}(z_1)}{\text{erf}(z_2) - \text{erf}(z_1)} \right]$$

4.4 The Fundamental Scaling Law: $\frac{x^2}{Dt} = \text{Constant}$

Notice that in Fick's Second Law solution, the ratio $\frac{C_x - C_0}{C_s - C_0}$ depends solely on the dimensionless argument:

$$z = \frac{x}{2\sqrt{Dt}}$$

- **The Constant-Composition Law:** For a given alloy system, to achieve the **exact same concentration** C_x under different processing conditions:

$$z = \text{constant} \iff \frac{x}{2\sqrt{Dt}} = \text{constant} \iff \frac{x^2}{D \cdot t} = \text{constant}$$

- **Case 1: Same Temperature ($D_1 = D_2 = D$), Changing Time and Depth:**

$$\frac{x_1^2}{t_1} = \frac{x_2^2}{t_2} \iff \frac{x_1}{\sqrt{t_1}} = \frac{x_2}{\sqrt{t_2}}$$

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Profound Engineering Consequence

: Diffusion depth scales with the **square root of time** ($x \propto \sqrt{t}$). To double the carburization depth ($x_2 = 2x_1$), the required furnace time **quadruples**: $t_2 = 4t_1$!

- **Case 2: Changing Temperature to Reduce Processing Time:**

$$D_1 t_1 = D_2 t_2 \implies t_2 = t_1 \left(\frac{D_1}{D_2} \right)$$

Increasing furnace temperature from 900°C to 1000°C drastically increases D , cutting required carburizing time from 20 hours to under 4 hours!

5. Factors Influencing Diffusion: Temperature & The Arrhenius Law (Callister §5.5)

5.1 Temperature Dependence: The Arrhenius Equation

Thermal energy facilitates atomic jumps over potential barriers. The diffusion coefficient D obeys an exponential **Arrhenius relationship**:

$$D = D_0 \exp \left(-\frac{Q_d}{RT} \right)$$

where:

- D = diffusion coefficient at temperature T (m²/s).
- D_0 = temperature-independent pre-exponential frequency factor (m²/s).
- Q_d = activation energy for diffusion (J/mol or eV/atom).
- R = universal gas constant = 8.314 J/(mol · K).
- T = absolute temperature in **Kelvin (K)**.

Linear Form for Experimental Data Fitting:

Taking the natural logarithm of both sides:

$$\ln D = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T} \right)$$

- A plot of $\ln D$ (vertical axis) versus $\frac{1}{T}$ (horizontal axis) produces a **straight line**:
 - Slope = $-\frac{Q_d}{R}$
 - Vertical y-intercept = $\ln D_0$
- Given diffusion coefficients D_1 and D_2 at two different temperatures T_1 and T_2 :

$$\ln \left(\frac{D_1}{D_2} \right) = -\frac{Q_d}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \implies Q_d = -\frac{R \ln(D_1/D_2)}{\frac{1}{T_1} - \frac{1}{T_2}}$$

5.2 Diffusion Pathways: Surface vs. Boundary vs. Volume

Atoms can migrate through different structural regions of a polycrystalline solid:

1. **Surface Diffusion:** Atoms migrate along external surfaces. Very low activation energy because surface atoms have few confining neighbor bonds.
2. **Grain Boundary Diffusion:** Atoms migrate along disordered grain boundary interfaces.
3. **Volume (Lattice / Bulk) Diffusion:** Atoms migrate directly through the interior of crystalline grains.

$$\text{Diffusion Rates: } D_{\text{surface}} > D_{\text{grain boundary}} \gg D_{\text{lattice}}$$

$$\text{Activation Energies: } Q_{\text{surface}} < Q_{\text{grain boundary}} \ll Q_{\text{lattice}}$$

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Temperature Crossover

: At low temperatures ($T < 0.5T_m$), grain boundary diffusion dominates because bulk atoms lack the thermal energy to overcome Q_{lattice} . At high temperatures ($T > 0.7T_m$), volume diffusion dominates overall mass transport because the total volume of grains is hundreds of thousands of times greater than the volume of grain boundary interfaces.

6. Comprehensive Step-by-Step Problem Walkthroughs

6.1 Problem 1: Industrial Gas Carburization of a Steel Gear

Problem Statement: A gear manufactured from a low-carbon steel containing an initial uniform carbon concentration of $C_0 = 0.20$ wt is placed in a carburizing gas atmosphere maintained at a constant surface concentration of $C_s = 1.20$ wt at a temperature of 950°C (1223 K).

1. Calculate the diffusion coefficient D of carbon in FCC γ -iron at 950°C , given pre-exponential factor $D_0 = 2.3 \times 10^{-5} \text{ m}^2/\text{s}$ and activation energy $Q_d = 148 \text{ kJ/mol}$ (148,000 J/mol).
2. Determine the time (in hours) required to achieve a carbon concentration of $C_x = 0.60$ wt at a depth of 0.75 mm ($7.5 \times 10^{-4} \text{ m}$) beneath the surface.

Step 1: Calculate the Diffusion Coefficient D at 950°C (1223 K)

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right) = (2.3 \times 10^{-5} \text{ m}^2/\text{s}) \exp\left(-\frac{148,000 \text{ J/mol}}{(8.314 \text{ J/mol} \cdot \text{K}) \times (1223 \text{ K})}\right)$$

Evaluate the exponent:

$$\frac{148,000}{8.314 \times 1223} = \frac{148,000}{10168.0} = 14.555$$

$$D = (2.3 \times 10^{-5}) \exp(-14.555) = (2.3 \times 10^{-5}) \times (4.773 \times 10^{-7}) = 1.098 \times 10^{-11} \text{ m}^2/\text{s}$$

Step 2: Set Up the Nonsteady-State Diffusion Equation

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

Substitute the known concentration values:

$$\frac{0.60 - 0.20}{1.20 - 0.20} = \frac{0.40}{1.00} = 0.4000 = 1 - \operatorname{erf}(z)$$

Rearrange to isolate the error function:

$$\operatorname{erf}(z) = 1 - 0.4000 = 0.6000$$

Step 3: Determine z via Linear Interpolation from Error Function Table

Look up $\text{erf}(z) = 0.6000$ in the table:

- At $z_1 = 0.55$: $\text{erf}(z_1) = 0.5633$
- At $z_2 = 0.60$: $\text{erf}(z_2) = 0.6039$ Since $0.5633 < 0.6000 < 0.6039$, apply linear interpolation:

$$\frac{z - 0.55}{0.60 - 0.55} = \frac{0.6000 - 0.5633}{0.6039 - 0.5633}$$
$$\frac{z - 0.55}{0.05} = \frac{0.0367}{0.0406} = 0.9039$$

$$z = 0.55 + (0.05 \times 0.9039) = 0.55 + 0.0452 = 0.5952$$

Step 4: Solve for Diffusion Time t

Recall that $z = \frac{x}{2\sqrt{Dt}}$:

$$0.5952 = \frac{x}{2\sqrt{Dt}} \implies 2\sqrt{Dt} = \frac{x}{0.5952} \implies \sqrt{Dt} = \frac{x}{2 \times 0.5952} = \frac{x}{1.1904}$$

Square both sides:

$$Dt = \left(\frac{x}{1.1904} \right)^2 \implies t = \frac{x^2}{(1.1904)^2 \cdot D}$$

Substitute $x = 0.75 \text{ mm} = 7.5 \times 10^{-4} \text{ m}$ and $D = 1.098 \times 10^{-11} \text{ m}^2/\text{s}$:

$$x^2 = (7.5 \times 10^{-4} \text{ m})^2 = 5.625 \times 10^{-7} \text{ m}^2$$

$$(1.1904)^2 = 1.4171$$

$$t = \frac{5.625 \times 10^{-7}}{1.4171 \times (1.098 \times 10^{-11} \text{ m}^2/\text{s})} = \frac{5.625 \times 10^{-7}}{1.556 \times 10^{-11}} = 36,150 \text{ seconds}$$

Convert seconds to hours:

$$t = \frac{36,150 \text{ s}}{3600 \text{ s/hr}} = 10.04 \text{ hours} \approx 10 \text{ hours}$$

6.2 Problem 2: Determining Activation Energy Q_d and Pre-Exponential D_0

Problem Statement: The diffusion coefficient for diffusion of zinc in copper was experimentally measured at two temperatures:

- At $T_1 = 600^\circ\text{C}$ (873 K): $D_1 = 5.6 \times 10^{-16} \text{ m}^2/\text{s}$
- At $T_2 = 800^\circ\text{C}$ (1073 K): $D_2 = 1.8 \times 10^{-13} \text{ m}^2/\text{s}$

- Calculate the activation energy Q_d (in kJ/mol).
- Determine the pre-exponential factor D_0 (in m^2/s).
- Predict the diffusion coefficient at $T_3 = 900^\circ\text{C}$ (1173 K).

Step 1: Calculate Activation Energy Q_d

Using the two-point Arrhenius formula:

$$\ln\left(\frac{D_2}{D_1}\right) = -\frac{Q_d}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) = \frac{Q_d}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

- $\ln\left(\frac{1.8 \times 10^{-13}}{5.6 \times 10^{-16}}\right) = \ln(321.43) = 5.7728$
- $\frac{1}{T_1} = \frac{1}{873} = 1.1455 \times 10^{-3} \text{ K}^{-1}$
- $\frac{1}{T_2} = \frac{1}{1073} = 0.9320 \times 10^{-3} \text{ K}^{-1}$
- $\frac{1}{T_1} - \frac{1}{T_2} = (1.1455 - 0.9320) \times 10^{-3} = 2.135 \times 10^{-4} \text{ K}^{-1}$ Solve for Q_d :

$$Q_d = \frac{R \ln(D_2/D_1)}{\frac{1}{T_1} - \frac{1}{T_2}} = \frac{(8.314 \text{ J/mol} \cdot \text{K}) \times 5.7728}{2.135 \times 10^{-4} \text{ K}^{-1}} = \frac{47.995}{2.135 \times 10^{-4}} = 224,800 \text{ J/mol} \approx 225 \text{ kJ/mol}$$

Step 2: Calculate Pre-Exponential Factor D_0

Using data at $T_2 = 1073 \text{ K}$:

$$D_2 = D_0 \exp\left(-\frac{Q_d}{RT_2}\right) \Rightarrow D_0 = D_2 \exp\left(\frac{Q_d}{RT_2}\right)$$

$$\frac{Q_d}{RT_2} = \frac{224,800}{8.314 \times 1073} = \frac{224,800}{8920.9} = 25.199$$

$$D_0 = (1.8 \times 10^{-13} \text{ m}^2/\text{s}) \times \exp(25.199) = (1.8 \times 10^{-13}) \times (8.786 \times 10^{10}) = 1.58 \times 10^{-2} \text{ m}^2/\text{s}$$

Step 3: Predict Diffusion Coefficient at 900°C (1173 K)

$$\frac{Q_d}{RT_3} = \frac{224,800}{8.314 \times 1173} = \frac{224,800}{9752.3} = 23.051$$

$$D_3 = D_0 \exp(-23.051) = (1.58 \times 10^{-2}) \times (9.752 \times 10^{-11}) = 1.54 \times 10^{-12} \text{ m}^2/\text{s}$$

7. Critical Concordia Exam Traps & Red Flags

- **⚠️ Trap 1: Forgetting the $1 - \text{erf}(z)$ Complement:** The formula is $\frac{C_x - C_0}{C_s - C_0} = 1 - \text{erf}(z)$. Students routinely forget the $1 - \dots$ and set $\frac{C_x - C_0}{C_s - C_0} = \text{erf}(z)$. In Problem 1, this would give $\text{erf}(z) = 0.4000$ instead of 0.6000 , resulting in a completely erroneous time calculation!
- **⚠️ Trap 2: Converting Hours to Seconds:** In Fick's Second Law, the diffusion coefficient D has units of m^2/s . Therefore, time t in $\frac{x}{2\sqrt{Dt}}$ **MUST be converted to seconds!** If an exam problem specifies $t = 10$ hours, you must substitute $t = 10 \times 3600 = 36,000 \text{ s}$.
- **⚠️ Trap 3: Mixing Millimeters and Meters in Depth x :** Diffusion depths are typically specified in millimeters ($x = 0.5 \text{ mm}$), but D is in m^2/s . You must convert x to meters ($x = 5 \times 10^{-4} \text{ m}$) before squaring. Forgetting this results in an error of 10^6 ($1,000,000\times$) in calculated time!
- **⚠️ Trap 4: Doubling Depth Requires $4\times$ Time:** A classic multiple choice question asks: "If it takes 2 hours to achieve a depth of 0.5 mm , how long does it take to achieve 1.0 mm at the same temperature?" Students incorrectly answer 4 hours (linear). The correct answer is **8 hours** ($x^2/t = \text{const} \Rightarrow (2)^2 \times 2 = 8 \text{ hrs}$)!